Somanshu Mohapatra

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SKILLS

Languages: Experienced in C++, Python, Embedded C, and MATLAB for algorithm development and simulation.
Robotics: Applied Forward/Inverse Kinematics (DH Parameters), ROS, and Gazebo for path planning and SLAM.
Electronics: Configured Arduino UNO, Raspberry Pi, STM, and motor drivers utilizing UART, I2C, SPI, and CAN.
CAD & Design: Designed assemblies using SolidWorks and Fusion, applying FEA and FDM/SLA printing for prototyping.
Soft Skills: Effective Organizer, Team Management, Strong Interpersonal Skills, and logical Problem Solving.

TRAININGS

AutomateX: Mastering PLCs for Industrial Automation | *LPU* — Mar' 25
View Training Certificate

- Acquired comprehensive knowledge of PLC architecture, ladder logic programming, and industrial automation principles.
- Simulated automated industrial workflows using TwidoSuite to manage sensor inputs and heavy actuator outputs.
- Tested fail-safe mechanisms and sequence patterns to reinforce practical understanding of factory floor operations.

PROJECTS

Dynamic Audio Monitoring and Narration | *Raspberry Pi, Python, AssemblyAI API* — Apr' 25 – May' 25

- Developed a real-time transcription system for the deaf using Raspberry Pi to provide accessible communication.
- Integrated AssemblyAI API and WebSockets with an I2C SSD1306 OLED display utilizing multi-threading technology.
- Achieved 95%+ transcription accuracy and under 300ms latency, enabling practical, on-the-go conversational assistance.

3-DOF Robotic Arm with Real-Time Control | *MATLAB, SolidWorks, Arduino IDE, Bambu Studio* — Apr' 24

- Constructed a 3-DOF robotic arm utilizing 3D-printed parts to optimize the payload-to-weight ratio for automated tasks.
- Programmed NEMA steppers and Arduino Mega, executing forward and inverse kinematics through MATLAB-UART integration.
- Decreased trajectory error by 20% during real-time operations, ensuring precise and reliable object manipulation.

Smart Energy Meter | *ESP32, IOT Monitoring, HTML Webserver* — Jun' 23

- Designed an IOT-based energy monitoring system to track residential power consumption through a real-time browser interface.
- Configured ESP32 and WIFI modules with ZMPT101B and ZMCT103C sensors, transmitting data to an HTML webserver.
- Delivered voltage and current measurements with precise accuracy, facilitating efficient remote energy management.

CERTIFICATES

Data Base Management System | NPTEL — Aug' 25
Fusioncraft | Resolute Lab — Mar' 24

ACHIEVEMENTS

METHOD FOR ENHANCING FOOD ACCESSIBILITY, MONITORING FRESHNESS, AND REDUCING ENERGY LOSS IN CYLINDRICAL REFRIGERATION UNITS — Sep' 25
Somanshu Mohapatra, Dr. Sumit Shoor
Application No.: 202511085342 A

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology — Aug' 23 – Present
Robotics and Automation Engineering; CGPA: 8.38

Shri S.N. Siddeshwar Public School — Gurugram, Haryana
Intermediate, PCMB; Percentage: 85.6% — Mar' 21 – May' 23

Ryan International School — Gurugram, Haryana
Matriculation, Percentage: 81.6% — Mar' 20 – May' 21